
7 AI-Based Waste Classification in the Healthcare Industry

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INTRODUCTION

The collection, storage, categorization, and transportation of solid wastes are all part of the waste management process, especially in the healthcare industry. While recovering from a serious illness or surgery, diet is crucial. Based on the needs of the patient, all reputable hospitals offer a balanced diet that is nutritious and healthy. For patients who are hospitalized for short or long stays, the hospital catering services are essential in supplying appropriate diet food. Hospital catering provides a great deal of vegetable wastes that are generated every day. People drink water in the bottle and throw the bottles in the dustbin. Also, everyday a huge amount of biological waste is generated in the healthcare industry. All these types of waste are all included in waste management. Planning, instruction, and training are essential for each of these processes. The waste is primarily produced by human activity. Glass, paper, plastics, wood, and vegetable waste are among the most typical types of waste. A significant role is played by waste classification, which is a time-consuming procedure, particularly in larger cities. Waste management varies greatly all over the world, with each country having its own approach; it is vital for living in a sustainable manner. A clean, hygienic, and pleasant environment necessitates proper waste management. People are currently engaged to manually segregate waste (hand-picking method) [1]. This necessitates a tremendous amount of human effort. Workers are at risk because they physically segregate garbage, exposing themselves to the risk of infection from germs found in the solid waste. It is prohibitively expensive to separate waste the traditional way. Municipal solid waste (MSW) is typically disposed of in landfills and bodies of water. Groundwater and surface waters are both contaminated by garbage dumped in landfills. In crowded cities especially, underground water is a crucial source of water for people, putting them in danger when it becomes contaminated. Batteries and lead materials at the landfill primarily cause soil pollution. According to estimates, there are roughly 10,000 metric tons of MSW produced annually. The environment is at risk if these wastes are not properly disposed of and separated. As soon as possible, waste should be separated into different

categories to increase the proportion of recycled and reused materials and decrease litter contamination. By separating waste, disposal costs are decreased. Additionally, waste segmentation enables us to use the best waste treatment and disposal practices. Incorrect garbage disposal, particularly with regard to common household waste, is to blame for about 75% of infectious diseases. Mixing of waste with used water from households and other elements led to the spread of typhoid, cholera, and tuberculosis. On the other hand, the air may become contaminated when these mixed/non-segregated wastes are burned. In the worst-case scenario, cancer could result from incineration of these wastes if they contain any hazardous materials [2]. This is mainly because when these wastes are burned, tiny harmful particles are released. Deep learning and computer vision are fields that are rapidly developing, especially in the area of waste management. Furthermore, as image processing techniques and architectures advance quickly, we are forced to focus our efforts on meeting the growing demand for waste classification. To recognize and categorize the recorded image data, convolutional neural networks (CNNs) are used. Waste is divided into recognizable groups and types using support vector machines (SVMs). In contrast to the traditional segregation method, this method is significantly more accurate and efficient than hand-picking [3].

RESEARCH METHODOLOGY

ADVANTAGES AND ADVANCEMENTS OF AI IN IMAGE CLASSIFICATION

Artificial Intelligence (AI) is a digital mind that processes, evaluates, and examines the data that is fed into the system. AI aims to simulate a human lifestyle with better reasoning and perception. It proves more efficient to make the right decisions faster.

With the recent advancements in this field, along with image manipulation, AI would overtake manual efforts with better understanding of data and accuracy. Machine learning is used to process and analyze images and AI recognizes the size, shape, position, and color. It consists of various algorithms that are used for manipulation of the images. The major aspects of image manipulation include image analysis and image segmentation, neural networks, SVMs, and a Gaussian mixture model [4]. Initially, the image is captured with the help of a sensor. Further, the quality of image is enhanced and corruption, if any, is also removed to acquire the mathematical model basis. RGB processing can be used for color images. Similarly, morphological processing processes the image shape and structure. AI and ML together increase the speed and efficiency of the process when large data is fed into the system. This combined effort of AI and ML also decreases the manual effort that is required for processing every single data.

Computer vision deals with training of system to recognize and process the images in the real world. Neural networks and deep learning are the tools that are used for processing complex image models. There are various popular neural networks for complex image processing. The most common network type is the CNN. This is adapted for image processing combined with AI, and it is a class of deep

learning networks. The structure of CNNs is much similar to the neuron structure in the human brain. This multilayer network has various layers for image manipulation. The convolutional layer filters and abstracts the fed-in image. Pooling layers remove the corruptions in the image. Normalization layers are used to improve the network's performance. The last layer, fully connected layer, joins the previous levels to obtain input and produce output.

Various implementations of AI algorithm with deep learning and neural networks opens up a wide door for large technical advancements in the world.

Applications:

1. Fingerprint voice identification technology

Fingerprint identification technology deals with the uniqueness of the fingerprint. This technology mainly concentrates on the color fingerprint that allows us to identify a person's identity. The voice identification technology groups the voices on different age groups based on sound and information of the data.

2. Face recognition technology

Face recognition technology is a biometric identification system that relates the biometric facial pattern to identify a person's identity. This technology proves to be more accurate because its deals with pupil features that are highly unique.

3. Radiofrequency ID (RFID) and barcode recognition technology

RFID involves electromagnetic waves to track and transfer data through a transmitter and receiver antenna. A barcode identification system includes QR tag techniques.

WASTE CLASSIFICATION AND ITS IMPORTANCE

Domestic waste management is one of the most pressing concerns facing environmental governance today. Residents are the primary producers of household garbage, and the problem is exacerbated in places with high populations and rapid consumption.

People's increasingly fulfilling material lives generate a rising amount of domestic garbage, which comes in a variety of forms. Although this demonstrates that our living conditions are rising, the impact of waste on our lives is also increasing. The worse the problem gets, the more vital it becomes to figure out how to better deal with domestic garbage. One of the most effective approaches is a domestic garbage classification or sorting system.

Using a waste sorting system or waste categorization is the best strategy and best way to combat waste pollution. Garbage classification is now an essential step in a nation's development. By classifying garbage, people can learn how to preserve resources, use resources wisely, develop good living habits, and enhance their own quality of life. Effective waste sorting habits will help someone become more aware of environmental issues, understand the value of life's resources, and develop the habit of resource conservation.

Chemicals are present in some modern waste, some of which, if unidentified, will lead to an increase in human incidence. Even if you landfill the garbage far from your home and use the necessary isolation technology, it is difficult to prevent the penetration of harmful substances that will enter the entire ecosystem with the Earth's circulation. The health of people will eventually be impacted by plant and animal pollution of soil and water supplies.

Waste classification is the process of grouping different types of trash into distinct categories according to the impact that each type's use has on the environment, the makeup of the trash, and the needs of the treatment technologies that are currently in use. Garbage classification serves the following purposes:

1. Separate the kitchen wastes, which are primarily composed of perishable organic ingredients, to produce high-quality raw materials for garbage composting and high-quality organic fertilizers, which will help to improve soil fertility and decrease fertilizer use.
2. By designating waste as hazardous, it is possible to treat waste without harming the environment and lower the danger of air, water, and soil pollution. Hazardous waste is waste that has trace amounts of pathogenic microorganisms, organic pollutants, and heavy metals.
3. Increase the percentage of waste that is recycled, lower the demand for raw materials, and lower CO₂ emissions.

CLASSIFIERS IN PRACTICE

Waste products are currently classified using a variety of techniques. Based on the material's density, waste is categorized using x-rays. The waste is sorted according to object size in a trommel separator, which consists of a rotating drum with perforations. With this technique, the drum is used to separate smaller waste particles from larger waste particles and collect each type separately. Large-scale induction sorting is a common practice for waste classification. This method involves placing and moving the waste along a conveyor belt. The conveyor belt is equipped with a number of sensors that detect waste and categorize it as metal, wood, plastic, and other materials.

Pressured air jets installed along the conveyor belt are then used to move these wastes to the proper collector bin. We make use of the electromagnetic theory to classify waste. Eddy current separator is the name given to this type of segregator, which is typically used to separate ferrous from non-ferrous waste when dealing with metallic waste. Because this Eddy current separator is primarily used for waste classification in metallic-based industries, it is not recommended for non-profit use. Near-infrared (NIR) sensors are the other type of segregator and, as of now, are the most successful in both profit and non-profit applications [5].

This technique categorizes waste objects according to their reflectance characteristics. Of all the waste classification techniques, this one is the most successful [6]. The ability to identify and categorize differs greatly from other methods because various materials have different reflectance characteristics. The results, however, may be incredibly unpredictable if one type of waste is piled on top of another.

DATA COLLECTION

In the field of object categorization using images, the dataset is absolutely crucial. To achieve the desired results, it is crucial to draw the appropriate conclusions from the supported fact data mining. It is defined as the effective application of large datasets using mathematical and statistical techniques. Image processing frequently employs MATLAB and OpenCV. In the dataset, there are 470 images. The information was specifically gathered to take into account the many factors that affect waste sorting. Other factors include mixed waste, overlaying, uneven illumination, and others.

Preprocessing of images is carried out to accommodate these discrepancies. To carry out data mining, we employ SVMs [7, 8] in the suggested strategy. With SVMs, classification and regression problems can be resolved. Each feature and its associated value are displayed by an SVM at a different coordinate in an n -dimensional space. Hyperplanes are divided according to the multidimensional information they hold, and this serves as the basis for classification.

$$\min_{y,w,b} \frac{1}{2} \|w\|^2 \quad (7.1)$$

$$y^{(i)} (w^T x^{(i)} + b) \geq 1, i = 1 \dots m$$

We use linear SVM for the specified classification. The optimization of SVM is represented by Eq. (7.1), where y^i is the cost function, w^T is the weights, b is bias, and i is the sample range from 1 to m , because there are more features in the image dataset than observations. SVMs are particularly effective for higher dimensional data. SVM decreases memory usage during image pre-processing. The strength of the decision function is also enhanced by the use of various kernel functions.

YOLOv4

CNN supports the YOLO architecture. The CNN separates an image into regions before predicting the boundary boxes and probabilities for each region. It forecasts a number of bounding boxes and probabilities for each class at the same time. YOLO implicitly encodes context in addition to visual information about classes because it sees the entire image during training and testing.

The majority of current accurate models are extremely slow and inefficient when trained with a small mini-batch size on a single GPU, and require multiple GPUs. By developing an object detector that can be trained on a single GPU with a smaller mini-batch size, this problem is addressed in YOLOv4. This enables the training of an extremely quick and precise object detector on a single 1080 Ti or 2080 Ti GPU.

Some backbones are better suited to classification than detection. For instance, CSPResNext50 outperformed CSPDarknet53 in terms of image categorization, but CSPDarknet53 outperformed CSPResNext50 in terms of object detection. According to the article, a backbone model for object detection needs more layers for a larger receptive field and a larger input network size for better identification of small objects. An extractor of features, A backbone is a deep learning framework.

Every backbone model is essentially a classification model. I'm going to assume that everyone at least knows about VGG16, one of the earliest deep learning classification systems. SqueezeNet, MobileNet, and ShuffleNet are three additional backbone models in addition to the ones mentioned earlier.

A subset of the bag of specialties called neck is made up of feature maps from different backbone phases; simply put, it is a feature aggregator. The neck of the object detection pipeline will be more thoroughly discussed in the following subsection.

The object detector, also referred to as the head, locates a potential object location but cannot identify the specific object that may be present. There are anchor-based and anchor-free detectors as well as two-stage and one-stage detectors available. The head will be covered in more detail in the sections that follow. YOLOv4 applies SAT to the training set, perturbing the data by an amount X until the predicted label matches the original class. This assists in the model's generalization.

The recognition of handwritten, printed, or typed characters from an image is known as optical character recognition (OCR). These methods are used to convert printed books to digital files. Data entry, traffic sign recognition, and other applications are examples.

Self-driving cars are vehicles that can drive themselves. Detecting pedestrians, automobiles, trucks, traffic signs, and other objects some of self-driving cars' most important capabilities. These detections are necessary for self-driving automobiles to function properly.

In iPhone and Android phones, face and IRIS verification and authentication are employed. If an exact face or iris match is detected, the device is authorized.

Object detection has a wide range of applications in robotics. Bin selecting and sorting of objects is one of the most prevalent applications. The robot can understand the location of objects using object detection algorithms. The robot will be able to pick up and sort the object using this information.

You can monitor an object and utilize it as an object counter by using object detection algorithms. For instance, how many automobiles have crossed at a crosswalk, how many people have entered a retail mall, and so forth

EXPERIMENT AND ANALYSIS

The proposed system consists of two compartments for biodegradable and non-biodegradable wastes, a servo motor, a disc for auxiliary lid rotation, a Raspberry Pi, a Raspberry Pi high-quality camera, and all of these parts put together. Additionally, a chamber for quick composting of biodegradable wastes has been built beneath this structure.

The suggested system includes a Pi, a high-resolution Pi camera, a motor, a disc for secondary closure rotation, and dual sections for recyclable and non-recyclable waste are included. Furthermore, a chamber beneath this structure is constructed for the fast composting of biodegradable wastes. Once dumped into the bin, the waste drops to the secondary lid below. The Raspberry Pi's high-resolution camera then captures the waste in low illuminance due to its interchangeable lenses that can acquire high-quality image data even in dark conditions. This image is then analyzed and classified by YOLOv4 weights that have been trained using explicit

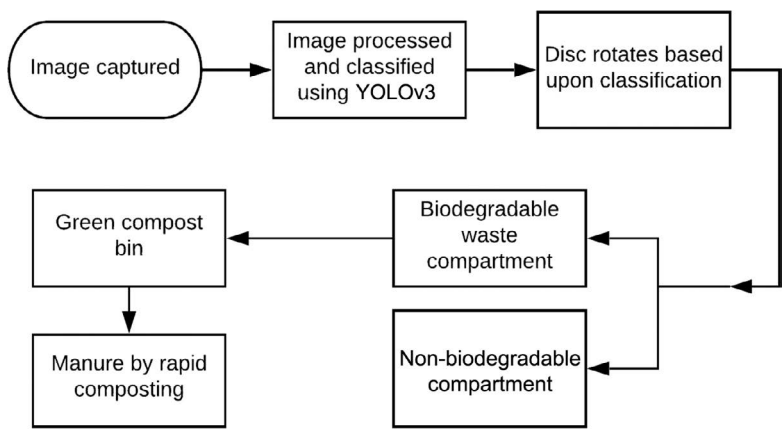


FIGURE 7.1 Flow of control and data in the proposed system.

data [9–14], and based on their classes, the disc rotates the auxiliary cover to direct it to its respective compartment. Afterward, biodegradable waste is shifted to a chamber devoted to quick composting according to Berkeley’s quick composting method [15–19], which involves thermal composting at a temperature of 55–65°C to maintain C:N balance at 30:1. Activators such as animal urine, nettles, old compost, or other natural activators are used for the decomposition process, which can take up to 14 days before converting waste into manure [20]. Furthermore machine learning algorithms such as SVMs are used for preprocessed image classification with the YOLOv4 technique with 470 photos divided into two categories (biodegradable and non-biodegradable). The dataset is labeled with LABELIMG, whereas Google Colaboratory is used for training with the maximum batch size set at 2500 and mean average precision (MAP) determined every 100 iterations.

Garbage that is thrown into the bin travels through the primary opening and under the secondary lid. The high-resolution camera on the Raspberry Pi then records the trash. Even in low-light we can record high-resolution image situations thanks to this camera’s interchangeable lenses. The control flow is given in below [Figure 7.1](#).

RESULTS

The results are significantly more accurate and efficient compared with training the YOLOv4 algorithm on the data without any picture preprocessing. In addition to increasing detection accuracy, image preprocessing with SMVs reduces memory usage. The percentages of organic material, carbon (C), nitrogen (N), and ash content decomposition after 14 days are displayed in [Table 7.1](#). This novel technique generates more nitrogen and carbon content when compared with cold composting. The process of removing the weeds from the garden was a success.

The precision was able to increase the effectiveness of waste classification in the suggested system by utilizing deep learning, and computer vision could be observed in [Figure 7.2](#). In addition to lowering the cost, this lessens the need for manual labor.

TABLE 7.1
Composted Contents and Percentages

S. No	Contents	Percentage (%)
1	Organic material	81.01
2	Carbon (C)	30.72
3	Ash	30.715
4	Nitrogen (N)	1.39

The outputs are given in [Figure 7.3](#). [Figure 7.4](#) represents the structure of the proposed bin structure. By lowering the size necessary for categorizing and decomposing biodegradable waste, the special composting of waste based on the Berkeley rapid composting technique achieves better results than the standard method. [Table 7.2](#) lists the components as referenced in [Figure 7.4](#), and [Table 7.3](#) shows the different components of the system along with their specifications.

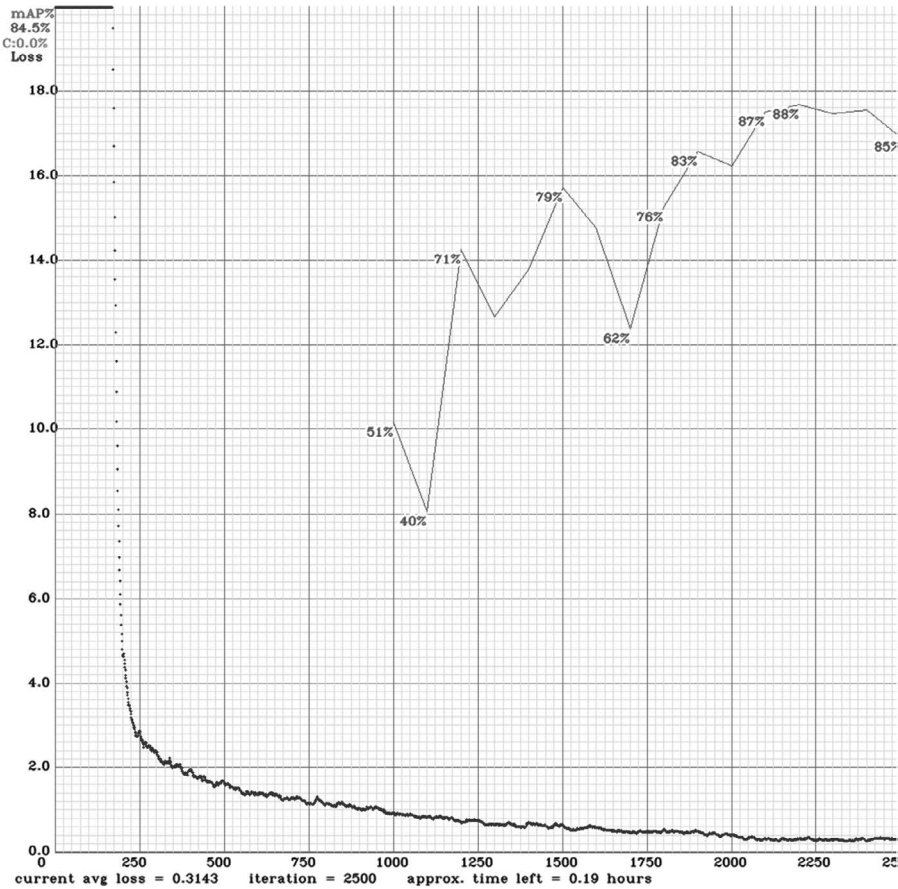


FIGURE 7.2 Mean average precision for 2500 epoch.

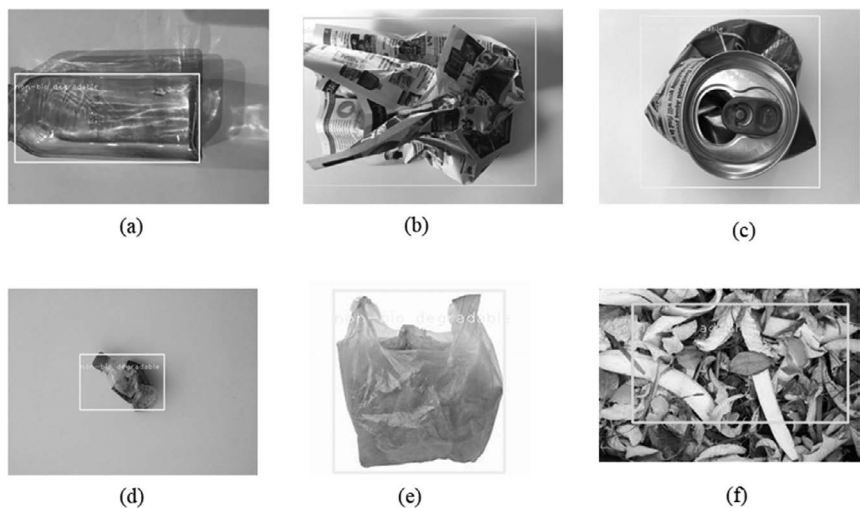


FIGURE 7.3 Model detection of biodegradable (b,f) and non-biodegradable (a,c,d,e) waste classified by the trained model.

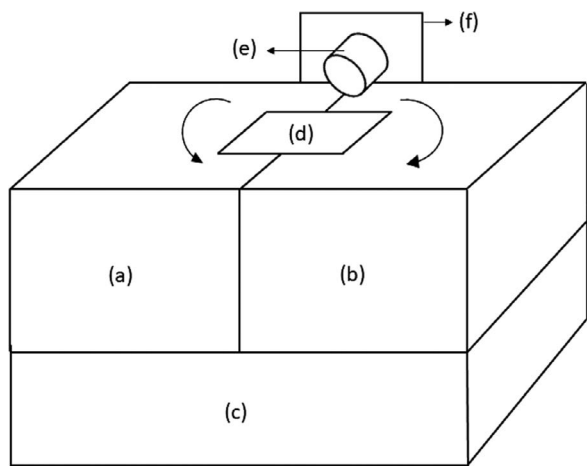


FIGURE 7.4 Structure of the proposed waste classifier and biocomposting system.

TABLE 7.2
Component as Referenced in Figure 7.4

Notations	Component
a	Biodegradable compartment
b	Non-biodegradable compartment
c	Biocomposting compartment
d	Waste in and flipping separator
e	Camera module
f	Processing module enclosing camera module

TABLE 7.3
Component and Specifications

Component Name	Specification	Number
Raspberry Pi 4	4GB RAM – 64bit, Cortex-A72Quad-core SoC @ 1.5GHz	1
Raspberry Pi camera	5MP Omnivision 5647 camera module	1
Servo motor	+5V typically – operating voltage T: 2.5kg/cm Speed:0.1 s/60° Rotation degree: 0°–180°	1
Battery	12V, 5Ah power safe battery	1

CONCLUSION

This chapter discusses the application of waste classification in the health industry with the help of AI. The proposal used machine learning and deep learning to solve the classification problem. The YOLO algorithm is utilized to classify the waste fed into the bin and SVMs were used to preprocess the image dataset used to train the algorithm. A waste composter based on Berkeley rapid composting was also attached to the unit. The works highlighted here primarily focus on waste classification to remove the mixture of waste that causes harm to both the environment and the ecosystem.

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